

Graph Neural Networks and BERT for Multimodal Music Context Understanding

A four-task study, with the negative results kept in

CSE425 — Neural Networks
Project Report

Abstract

We implement and evaluate a four-task pipeline for music context understanding: (1) a DistilBERT caption encoder predicting a 50-tag vocabulary, (2) a graph neural network over an audio-segment graph for genre classification, (3) a cross-attention fusion model predicting tags and continuous valence/arousal jointly, and (4) a contrastive audio-text embedding space. All four run on real data: 929 GTZAN tracks, 1,802 DEAM excerpts, and 1,657 MusicCaps clips whose audio we reconstructed from source ourselves (a 30.0% recovery rate, accounted for in Section 3.3). Every number reported here is measured, and every claim is stated against a control. The controls are what make the study informative, because three of them contradict the hypothesis the pipeline was built to support. A plain CNN on log-mel spectrograms reaches **0.5683** macro-F1 on GTZAN, beating the best graph model (0.4999) by 0.0684; deleting the temporal edges *improves* the GNN by 0.1588 macro-F1, and deleting all edges entirely costs nothing measurable; and on raw MusicCaps captions a regular expression scores 0.5581 macro-F1 against DistilBERT’s 0.5933, so most of the apparent “language understanding” in the naive setup is substring matching. Masking tag surface forms from the captions removes the shortcut and costs the model 0.1241 macro-F1, which we take as the size of the leak. We report what the graph formulation does and does not buy, and where the honest bottleneck is data rather than architecture.

1 Introduction

“Music context” in this project means the union of three things a listener infers effortlessly and a model does not: what genre a recording belongs to, what mood it carries, and what a person would say about it in a sentence. These are usually studied separately, with separate architectures and separate corpora. The premise of this project is that they are the same problem viewed through three channels, and that a model with access to more than one channel should do better than a model with access to one.

The specific architectural bet is that *a recording is a structured object, not a bag of frames*. A song restates its chorus; a jazz solo returns to the head; a build-up and its release are the same material at different intensities. A representation that can see those relations — a graph whose nodes are segments and whose edges say “these two stretches are the same music” — should therefore be strictly more informative than a flat pooled summary. The second bet is that a language encoder and a graph encoder can be joined, either by cross-attention (Task 3) or by a contrastive objective (Task 4), into something that behaves like a shared semantic space.

This report tests both bets rather than assuming them. Our contributions:

- **A complete, runnable four-task implementation** over four corpora, including audio we had to reconstruct ourselves because MusicCaps ships captions and YouTube IDs but no audio (Section 3.3).
- **A control for every claim.** A lexical-match baseline for the text task, a random-pair control for the graph-coherence statistic, explicit chance rates (K/n) for retrieval, and the training-mean predictor as the floor for emotion regression. Several results collapse against their control, and we report those.
- **Three negative results** that argue against the project’s own hypothesis: a mel-CNN beats every graph model on genre; the segment graph’s temporal edges are a net cost, and its full edge set buys nothing over no graph at all; and the text task as originally specified is largely solvable by substring matching.
- **Reproducible artefacts:** 83 exported example graphs, 24 generated figures, two executed notebooks, and a numbers layer that makes it impossible for the prose here to drift from the runs.

2 Related Work

Graph learning on audio. GraphSAGE [6] and GAT [17] are the two message-passing operators we use; both aggregate over a node’s neighbourhood, GAT with learned attention weights. Applying them to audio

Corpus (graph)	Graphs	Mean nodes	Mean edges	Mean degree	Node dim
gtzan (segment)	929	18.0000	50.9925	2.8329	340
deam (segment)	1802	18.0000	49.4184	2.7455	340
musiccaps (segment)	1507	12.0000	24.6689	2.0557	340
gtzan (chord)	929	5.1668	6.9989	1.3364	40

Table 1: Cached graph statistics per corpus.

requires choosing what a node is, and that choice is the substance of the modelling. We use fixed overlapping segments and, separately, a chord-transition graph. Known failure modes of deep message passing — in particular oversmoothing, where repeated averaging drives node states toward a common vector [9] — turn out to matter here (Section 6.3).

Language models for music description. MusicCaps [1] provides 5,521 expert-written captions for 10-second YouTube clips and is the only large corpus of free-text music description with aligned audio timestamps. We use DistilBERT [14], a 6-layer distillation of BERT [3], as the text encoder throughout.

Joint audio–text spaces. CLIP [12] established the recipe — two encoders, one InfoNCE objective [11], a learned temperature — and CLAP [4] and MuLan [7] carried it to audio and music respectively, at corpus scales three to five orders of magnitude larger than what we have. Our Task 4 is that recipe with a graph encoder in the audio branch and 591 training pairs; the result (Section 6.5) is mostly a statement about the second number.

Benchmarks. GTZAN [16] is the standard genre corpus and also a well-documented flawed one: Sturm [15] catalogued repetitions, mislabellings and artist duplication across its splits. We use the fault-filtered partition of Kereliuk et al. [8], which was constructed so that no artist appears on both sides. DEAM [2] supplies continuous valence/arousal annotations on Russell’s circumplex [13]. Implementation rests on PyTorch Geometric [5] and librosa [10].

3 Data

Four corpora, each carrying a different supervision signal, are cached as PyTorch Geometric graphs by a single preprocessing pass. Table 1 gives the resulting graph statistics; Figure 2 shows the distributions behind the means.

3.1 GTZAN (genre)

Ten genres, 100 thirty-second clips each. We use the fault-filtered partition [8]: 442 train / 197 validation / 290 test. The partition file lists 930 tracks; we build 929 graphs. The missing one is `jazz.00054`, which is truncated in the canonical archive and fails to decode (`libsndfile: format not recognised`) — a defect independent of this project and, we note, one of the faults that motivated the filtered partition in the first place. We report 290 test tracks throughout, so the numbers here are directly comparable to published fault-filtered results rather than to the inflated figures obtained on GTZAN’s original random splits.

3.2 DEAM (emotion)

1,802 excerpts with static valence and arousal ratings, split 1,261/270/271. Both targets are standardised to zero mean and unit variance on the training split, so the reported MAE is in standard deviations and a mean-predictor baseline sits near 1.0; the measured floors are 0.994 (valence) and 1.098 (arousal), and any regression that does not beat those is doing nothing. Valence and arousal are not independent in this corpus — their sample correlation is 0.57 over all 1,802 excerpts (Figure 1), which is worth remembering when reading the two MAE columns as though they were separate achievements.

3.3 MusicCaps, and what a 30% recovery rate means

MusicCaps distributes captions and YouTube IDs. Google does not redistribute the audio, so a project that wants *real* paired (audio, text) data has to reconstruct it. The alternative — templating captions from metadata labels — writes the label into the text channel and makes any fusion result circular, so we did not take it.

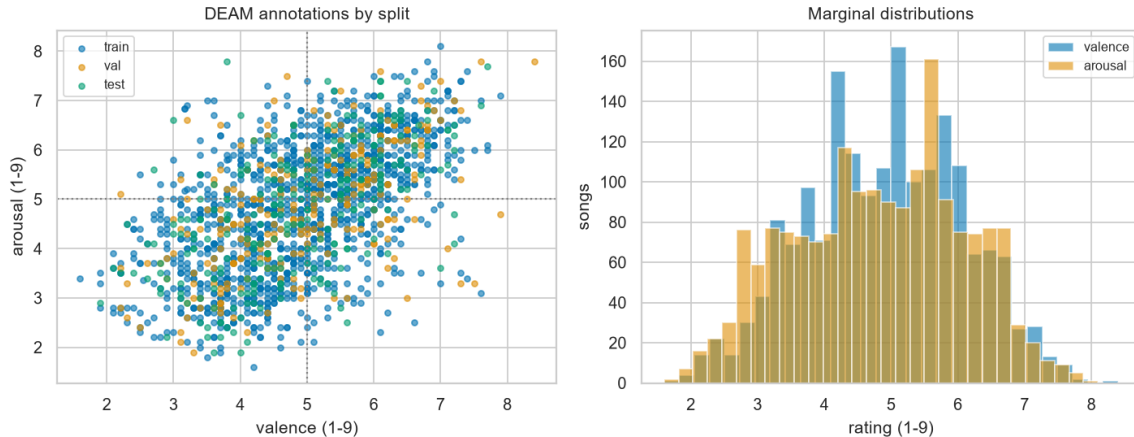


Figure 1: DEAM valence–arousal distribution. The two targets correlate at 0.57 across 1,802 excerpts, so the cloud runs along the low-valence/low-arousal to high-valence/high-arousal diagonal rather than filling Russell’s plane. A model that learns valence therefore gets much of arousal for free, and the two MAE columns in Table 5 should not be read as independent results.

Our downloader (`scripts/download_musiccaps_audio.py`) fetches the audio stream, decodes only the annotated $[t_{\text{start}}, t_{\text{end}}]$ window with PyAV, and writes a 22,050 Hz mono WAV. We recovered 1,657 of 5,521 nominal clips, or 30.0%, in 1.0 h of wall-clock. The 3,864 we did not get break down in a way that matters:

- 231 clips are genuinely gone — 141 removed or terminated, 80 set private, 10 region-blocked. No amount of retrying recovers these.
- 3,622 failed on YouTube’s anti-automation gate (“sign in to confirm you’re not a bot”). These clips exist; we simply could not fetch them unauthenticated. We retried each ID across six different player-client impersonations, which is what took the recovery rate from a first-pass figure to 30.0%, but the gate is the binding constraint.
- 11 failed for other reasons (silent or too-short decode window, network).

The distinction is the honest part: *only 231 clips are unavailable in principle*. The remaining shortfall is an access limitation of this build, not a property of the dataset, and it is recoverable by an authenticated session. We report every MusicCaps result on the clips we actually have, never on the nominal 5,521, and Task 4’s weakness (Section 6.5) is substantially a consequence of this number.

After keeping only clips with at least one in-vocabulary tag and successfully built graphs, the paired set is 1,507 graphs split 591/141/775. Note that test is *larger* than train. This is deliberate: MusicCaps publishes an `is_audioeval` subset, and we hold exactly that subset out as test rather than drawing our own split, so our test set is comparable to other work on this corpus. The cost is that recovery losses fall disproportionately on the training side, leaving 591 pairs to learn from. It is a defensible trade — a published test set is worth more than a balanced one — but it is a trade, and it caps what Task 4 can show.

3.4 The 50-tag vocabulary and its lexical shortcut

Tags are mined from MusicCaps’ `aspect_list` field: 13,219 distinct aspects reduced to the top 50 by frequency (*low quality, instrumental, emotional, noisy, passionate, ...*). Keeping only clips carrying at least one top-50 tag leaves 4,995 of 5,521. Support ranges from 147 to 1,217 clips, an $8.3\times$ imbalance (Figure 3), which is why macro-F1 rather than accuracy is the headline metric everywhere.

There is a problem with using these tags as targets for a caption encoder, and it is severe enough to change the experimental design. The aspects were written by the same annotators as the captions, so the tag string usually appears verbatim in the caption. We measured it three ways, tightening the condition each time. **4,671 of 4,995 captions (93.5%) contain a literal surface form of some vocabulary tag**, and 4,118 (82.4%) contain one of the tags the clip is actually labelled with — so for four clips in five, string matching alone yields a true positive with no learning involved. Dropping the top-50 truncation and asking only whether a caption echoes any of its own aspects puts the ceiling at 4,900 (98.1%). Matching is case-insensitive on whole-word boundaries, the same matcher the masking step deletes with, so these figures describe exactly the experiment reported in Section 6.1. A model given the raw caption can score well by pattern matching, and it does.

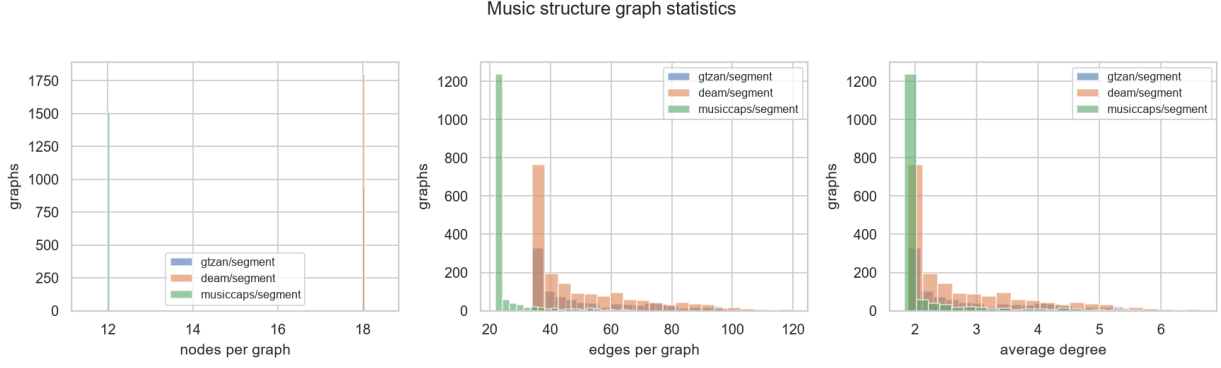


Figure 2: Graph statistics across corpora: node counts, edge counts, degree, and the temporal/similarity edge split. The segment graphs are deliberately sparse; see Section 4.2.1.

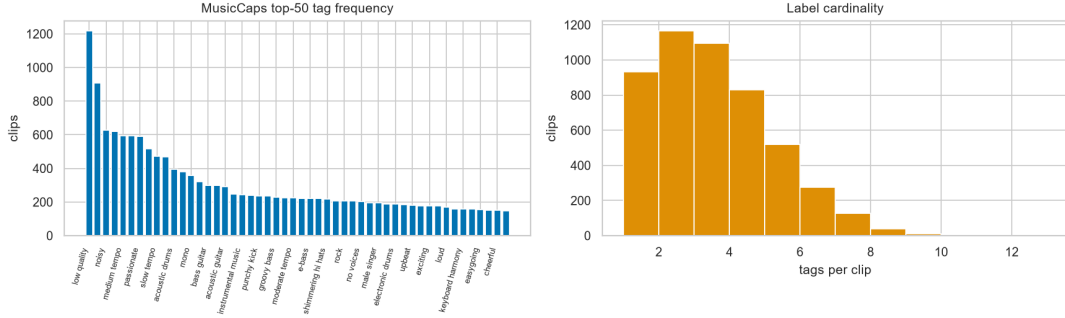


Figure 3: Support of the 50-tag MusicCaps vocabulary. The $8.3\times$ head-to-tail imbalance is why macro-F1 is the reported metric.

We therefore construct a *masked* caption variant in which every tag surface form is replaced with a sentinel, and treat the masked condition as the real task. The unmasked condition is retained only as a diagnostic — the gap between the two is our estimate of how much of the task is leakage.

4 Method

4.1 Audio front end

Audio is resampled to 22,050 Hz mono and normalised per track. We compute a 128-bin log-mel spectrogram, 12-bin chroma, and 20 MFCCs with a 2048-sample window and 512-sample hop. Per segment we take the mean, standard deviation, and first-difference statistics of each representation and concatenate them, giving a 340-dimensional node feature vector. Because node features are *per-segment statistics*, their dimensionality is independent of clip length, and graphs from different corpora remain mutually compatible — which is what lets Task 3 batch GTZAN and DEAM graphs together.

4.2 Segment graph

Nodes are fixed windows of 3.0 s with a 1.5 s hop (50% overlap). Two edge families connect them:

- **temporal** edges $i \rightarrow i+1$, encoding playback order;
- **similarity** edges between non-adjacent segments whose cosine similarity exceeds τ , encoding restatement — the chorus returning. At most **topk**= 4 per node, to keep the graph sparse.

All edges are made undirected. A 30 s GTZAN clip yields 18.0 nodes and 51.0 edges on average; MusicCaps’ 10 s clips use a per-corpus override (1.5 s window, 0.75 s hop) to reach 12.0 nodes rather than the 5 the default would give, since 5 nodes is too thin for 3 rounds of message passing to do anything.

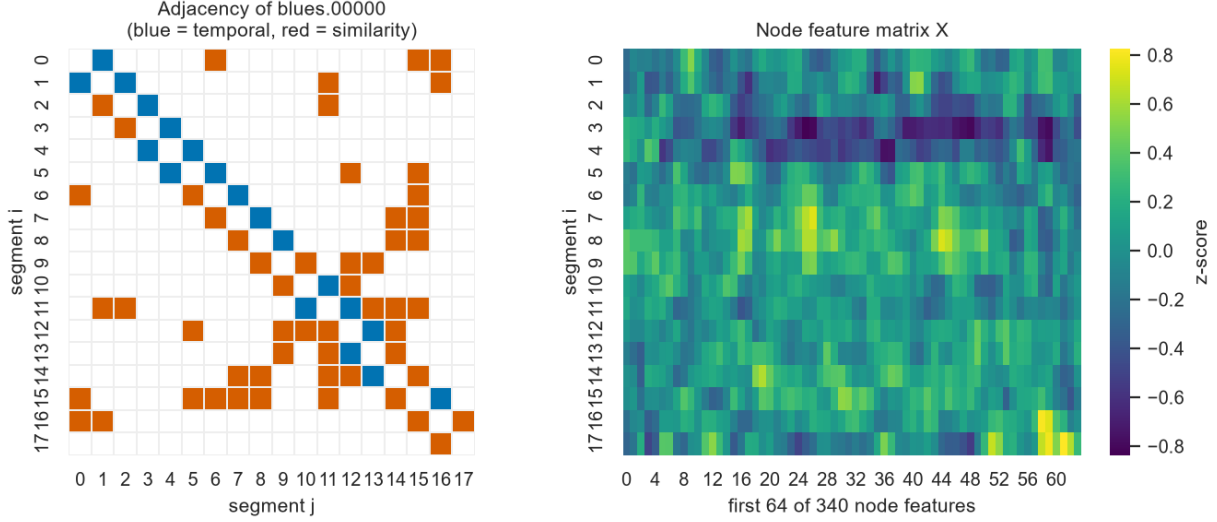


Figure 4: A segment graph and a chord-transition graph for the same track. Temporal edges form the backbone; similarity edges (the long-range arcs) mark restated material.

4.2.1 Calibrating τ

τ is not a free aesthetic choice, and setting it by intuition breaks the experiment. Segments of one song are intrinsically alike: over 2,720 non-adjacent pairs from 20 GTZAN tracks, the cosine distribution has $p_{25} = 0.716$, $p_{50} = 0.850$, $p_{90} = 0.961$, $p_{99} = 0.989$. An intuitive-looking $\tau = 0.85$ therefore admits 50.1% of *all* pairs, giving mean density 0.267; the graph becomes near-complete, message passing degenerates toward mean pooling, and the GNN becomes indistinguishable from the mean-pooled-features MLP baseline — the “structure helps” claim would be untestable rather than false. At $\tau = 0.95$ only 14.6% of pairs are admitted (mean density 0.175, average degree 2.98, and 60.9% of surviving edges temporal), retaining only genuinely repeated material while temporal edges keep the graph connected. We use $\tau = 0.95$ throughout and report the sweep that justifies it (`scripts/calibrate_tau.py`, `results/tau_calibration.json`).

4.3 Chord-transition graph

As a second, deliberately harsh structural view, we build a graph whose nodes are the 24 major/minor triads plus a no-chord symbol, obtained by template matching against the chroma vector, and whose edges are observed transitions weighted by count. This discards timbre entirely and keeps only harmonic motion. Averaging 5.2 nodes and 7.0 edges, it is a very lossy representation, and we include it to bound how much of genre is harmony alone.

4.4 Task 1 — caption $\rightarrow$ tags

Let X be a tokenised caption. We take the CLS vector of distilbert-base-uncased, $t = \text{BERT}_{\text{CLS}}(X) \in \mathbb{R}^{768}$, and predict each tag independently with a linear head,

$$\hat{y}_k = \sigma(w_k^\top t + b_k), \quad k = 1 \dots 50, \quad (1)$$

trained with summed binary cross-entropy. The encoder is fine-tuned with a discriminative learning rate (2×10^{-5} for BERT, 10^{-3} for the head), linear warmup over the first 10% of steps then cosine decay. The decision threshold is tuned on validation macro-F1, not fixed at 0.5; it settles at 0.700.

4.5 Task 2 — GNN genre classification

Node states are updated by GraphSAGE convolution,

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot \text{CONCAT} \left[h_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)} \right] \right), \quad (2)$$

for $l = 0 \dots 3-1$ with hidden width 256, batch normalisation, and dropout 0.3. A mean readout pools the graph, $g = \frac{1}{|V|} \sum_i h_i^{(L)}$, and a linear classifier maps g to ten genres. We ablate the operator (GraphSAGE vs.

GAT with 4 heads), the graph kind (segment vs. chord), and — most informatively — the edge set: temporal only, similarity only, and no edges at all. The last is the control that matters: with an empty edge set the model is a per-node MLP followed by mean pooling, so any gain the graph provides must show up as a gain over it.

4.6 Task 3 — cross-attention fusion

The graph gives $g \in \mathbb{R}^d$; the text encoder gives token states $H_{\text{text}} \in \mathbb{R}^{L \times d}$. We attend from the graph summary to the caption,

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad z = \text{CONCAT}(g, AH_{\text{text}}), \quad (3)$$

with Q projected from the graph side and K, V from the text side, 4 heads, $d = 256$. Two heads read z : a 50-way multi-label tag head and a 2-dimensional valence/arousal regressor. The loss is the specified combination

$$\mathcal{L} = \mathcal{L}_{\text{tags}} + \alpha \|v - \hat{v}\|^2 + \beta \|a - \hat{a}\|^2, \quad (4)$$

with $\alpha = 0.5$, $\beta = 0.5$. Because no single corpus carries both tags and valence/arousal, each batch mixes MusicCaps rows (tags, no emotion) with DEAM rows (emotion, no tags) and each loss term is masked to the rows that supply its label: 591 tagged and 1,261 emotion-labelled rows out of 1,852 training rows. We ablate the fusion operator over `{bert_only, gnn_only, concat, cross_attention}`, training four separate models. The first two are the single-modality controls, without which “fusion helps” is unfalsifiable.

4.7 Task 4 — contrastive alignment

Graph and text encoders are trained to agree via InfoNCE with a learnable temperature initialised at 0.07,

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(g_i, t_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(g_i, t_j)/\tau)}. \quad (5)$$

The specification’s loss is one-directional ($g \rightarrow t$); since the evaluation asks for recall in both directions we use the symmetric CLIP-style average of $g \rightarrow t$ and $t \rightarrow g$. Negatives come from the in-batch other pairs, so the batch size (`FourBatchSize`) is the number of negatives — a point we return to.

5 Experimental Setup

Everything runs on CPU (12 threads, no CUDA), which shaped the budget and is reported rather than hidden. Seeds are fixed at 425 for Python, NumPy and PyTorch. Optimisation is AdamW with weight decay 0.01 (transformer runs) or 0.0005 (GNN runs), cosine schedules, gradient clipping at 1.0, and early stopping on validation macro-F1. Text batches are padded dynamically to the longest sequence in the batch rather than to `max_length`, which is worth roughly a factor of two on CPU.

Two budget decisions are material to how the results should be read. Task 2 trains to 60 epochs with patience 15, so those models are converged or early-stopped on their own terms. Task 3 is capped at 6 epochs per ablation: one `cross_attention` epoch over 1,852 rows costs 3m 56s here, and the ablation list trains 4 models, so 30 epochs each would be ≈ 8 h. Task 3 numbers are consequently *lower bounds*, and Figure 11 is the evidence for whether 6 was enough: where validation loss is still falling at the last epoch, the model had not finished learning and we say so rather than presenting it as converged.

6 Results

6.1 Task 1: the text task is mostly substring matching

Table 2 gives the caption→tag results. Read the first two rows first, because they reframe the rest.

The lexical baseline — a regular expression, no learning, no parameters — gets 0.5581 macro-F1 on raw captions. DistilBERT fine-tuned on the same raw captions gets 0.5933. A 66.4-million-parameter language model fine-tuned for 6 epochs is therefore worth about 0.035 macro-F1 over `re.search`, which is a much less impressive claim than the raw number looks like in isolation, and it is the claim the naive version of this task actually supports.

Model	Macro-F1	Micro-F1	Samples-F1	AUC-PR	Thr
B1 random scores	0.1039	0.1078	0.1044	0.0619	0.5000
B1 training prior	0.0000	0.0000	0.0000	0.0598	0.3000
B0 lexical match (raw caption)	0.5581	0.5580	0.5146	0.4063	0.5000
B0 lexical match (masked caption)	0.0000	0.0000	0.0000	0.0598	0.5000
DistilBERT (raw caption)	0.5933	0.6297	0.5954	0.6111	0.7000
DistilBERT (masked caption)	0.4692	0.5320	0.4882	0.4618	0.7000

Table 2: Task 1: caption to multi-label tag prediction on the MusicCaps test split. The masked variant has every tag surface form removed from the caption.

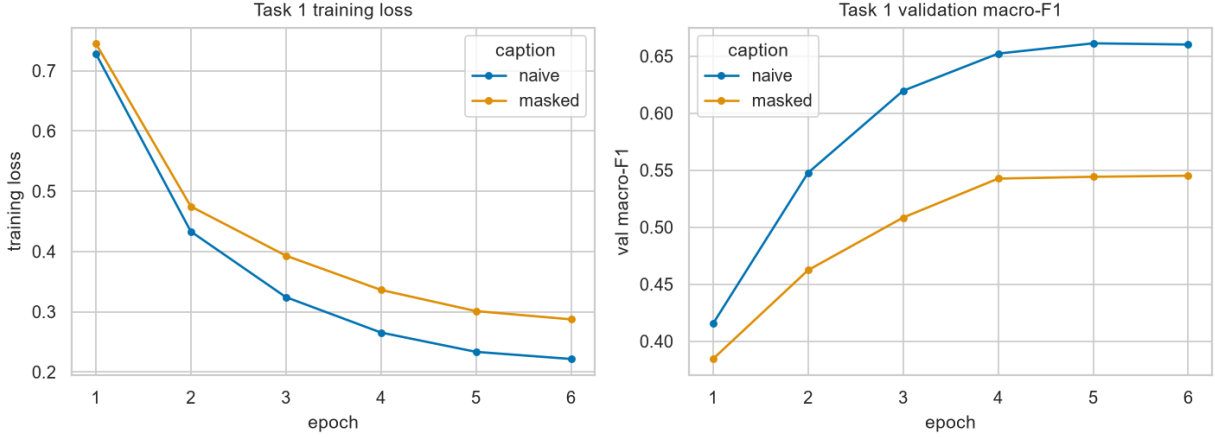


Figure 5: Task 1 learning curves, masked and naive conditions. The naive run tracks persistently higher on validation — the shortcut is visible during training, not only in the final metric.

Masking removes the shortcut cleanly. The same regex on masked captions scores 0.0000 — exactly zero, as it must, since the strings it looks for are gone. DistilBERT on masked captions scores **0.4692** macro-F1, 0.5320 micro-F1 and 0.4618 AUC-PR, well clear of the random-score baseline (0.1039) and of the training-prior baseline (0.0000, which predicts no tag at all above threshold and so scores nothing). That 0.4692 is the defensible measurement of caption understanding in this project: it is inference from surrounding context about a word that is not present.

The gap between conditions, 0.1241 macro-F1, is our estimate of the leak. We consider reporting it more useful than reporting either condition alone, and we would treat any published caption→aspect result on MusicCaps that does not control for this as uninterpretable.

6.2 Task 2: the graph is a net cost

Table 3 and Figure 7 give the genre results on the 290-track fault-filtered test split. Three findings, in increasing order of how much they cost the project’s hypothesis.

(i) **Removing the temporal edges improves the model.** The full segment graph reaches 0.3411 macro-F1. The similarity-only variant — same architecture, same features, temporal edges deleted — reaches **0.4999**, an improvement of 0.1588. The temporal-only variant is the worst of the three at 0.2994. The playback-order backbone, which is the intuitive part of the graph, is actively harmful. Section 6.3 gives our explanation: temporal edges are dense, they connect segments that are already highly correlated, and 3 rounds of averaging over them smooths node states toward a per-track mean.

(ii) **Removing *all* edges costs nothing.** With an empty edge set the model is a per-node MLP plus mean pooling, and it scores 0.3604 against the full graph’s 0.3411 — nominally ahead, by 0.0193. That margin is inside the $\lesssim 0.02$ band we decline to read from a single seed (Section 8), so the defensible claim is the weaker and still damaging one: message passing over the graph we built is worth no more than not passing messages at all, unless the edge set is restricted to similarity edges alone. The similarity-only margin over the full graph (0.1588) is an order of magnitude larger and does clear the band.

(iii) **A plain CNN beats every graph model.** B2, a four-block convolutional network on the log-mel spectrogram with 391,370 parameters, reaches **0.5759** accuracy and **0.5683** macro-F1 — 0.0684 macro-F1 above the best graph model, with fewer parameters than the GraphSAGE model’s 517,514. The segment-

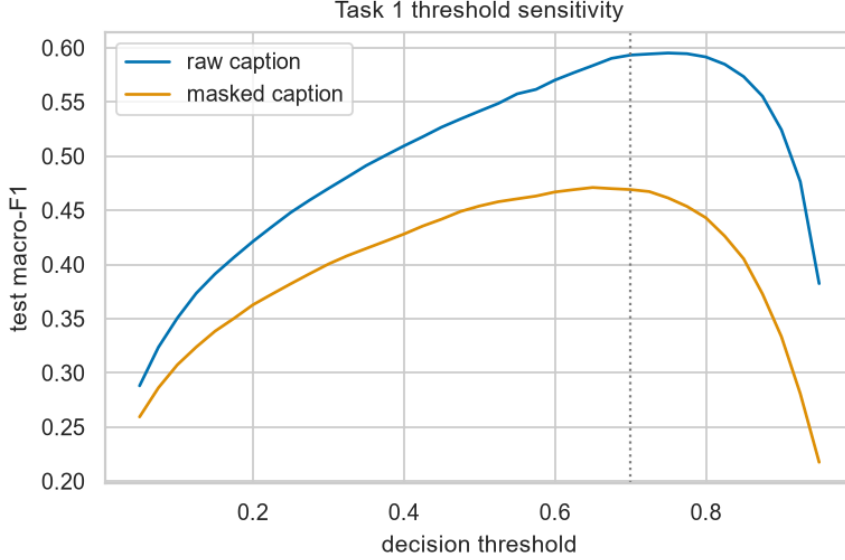


Figure 6: Macro-F1 against decision threshold. The optimum is at 0.700, not 0.5; tuning it on validation is worth a non-trivial amount of F1 on an imbalanced multi-label problem.

Model	Accuracy	Macro-F1	AUC-PR	Params (M)	Best epoch
B1 majority class	0.1069	0.0193	0.1000	0.0000	-1
B2 CNN on log-mel	0.5759	0.5683	0.6398	0.3914	19
B4 mean-pooled features + MLP	0.3138	0.2967	0.3139	0.1220	38
GraphSAGE, segment graph (ours)	0.3828	0.3411	0.4953	0.5175	9
GAT, segment graph	0.3483	0.2732	0.4283	0.9169	9
GraphSAGE, chord-transition graph	0.1655	0.1216	0.1604	0.4407	4
GraphSAGE, temporal edges only	0.3586	0.2994	0.4331	0.5175	15
GraphSAGE, similarity edges only	0.5034	0.4999	0.5449	0.5175	57
GraphSAGE, self-loops only	0.3931	0.3604	0.5149	0.5175	9

Table 3: Task 2: GTZAN genre classification on the fault-filtered test split (290 tracks). Parameter counts are trainable parameters.

statistics front end throws away exactly what a CNN exploits: local time–frequency texture. Summarising a 3.0s window by its per-band mean and variance discards the spectro-temporal detail that distinguishes a distorted guitar from a saturated synth pad, and no amount of message passing between segments puts it back.

The remaining ablations are consistent. GAT underperforms GraphSAGE (0.2732 vs. 0.3411) while using 916,874 parameters against 517,514—attention over a near-uniform neighbourhood buys little and costs capacity. The chord graph is weakest at 0.1216, above the majority-class floor (0.0193) but far below everything else: harmonic motion alone does not identify genre in GTZAN, which is unsurprising given that blues, jazz and rock share progressions. The mean-pooled-features MLP (B4, 0.2967) lands where a bag-of-segments model should.

6.3 Is the learned graph structure meaningful at all?

The specification proposes a graph-coherence statistic: the fraction of edges whose endpoint embeddings are similar,

$$S_{\text{graph}} = \frac{1}{|E|} \sum_{(i,j) \in E} \mathbf{1}[\cos(h_i, h_j) > \tau]. \quad (6)$$

Taken literally this number is uninformative, and it is worth saying why. After ReLU and batch normalisation the node states lie in the positive orthant, where almost any two vectors have positive cosine. At $\tau = 0.5$ we measure $S_{\text{graph}} = 0.9963$ — but a control that samples *random non-adjacent pairs* scores 0.5775 on the same embeddings. A statistic that assigns 0.5775 to random pairs is measuring the geometry of the activation

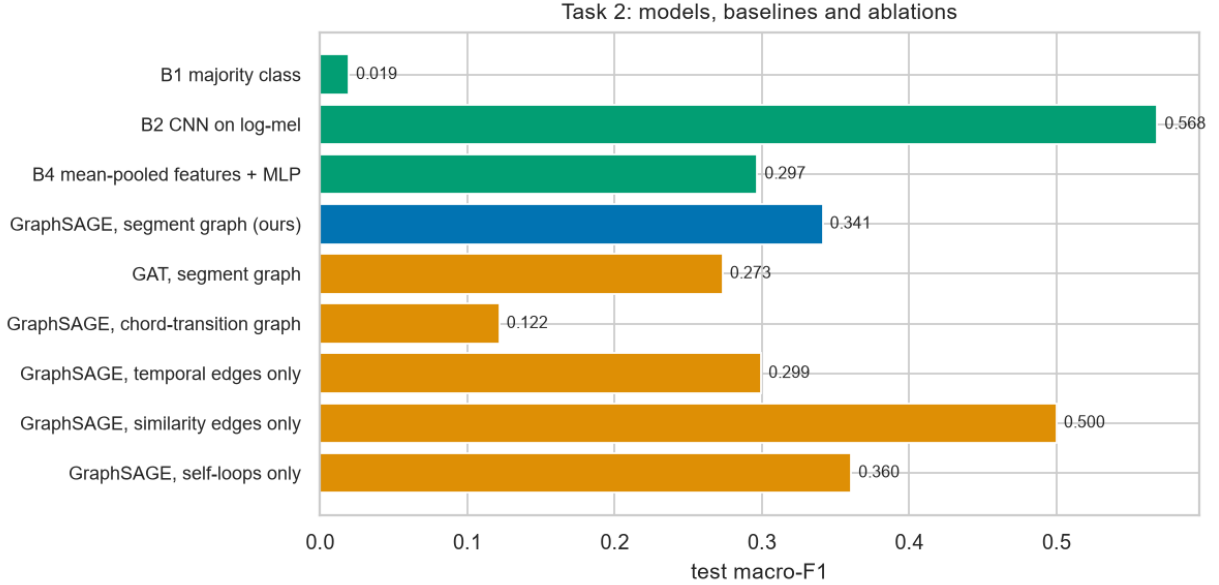


Figure 7: Task 2 ablations. Similarity-only and no-edges both beat the full segment graph; the mel-CNN baseline beats all of them.

tau	S_graph	Random pairs	Lift	Temporal edges	Similarity edges
0.5	0.9963	0.5775	0.4188	0.9939	0.9996
0.8	0.8489	0.0747	0.7742	0.7944	0.9361
0.9	0.5396	0.0123	0.5273	0.4472	0.6902
0.95	0.2531	0.0023	0.2508	0.1856	0.3694
0.98	0.0657	0.0002	0.0655	0.0483	0.0936
0.99	0.0168	0.0001	0.0167	0.0125	0.0224
0.995	0.0030	0.0000	0.0030	0.0024	0.0038
0.999	0.0000	0.0000	0.0000	0.0000	0.0000

Table 4: Graph coherence S_{graph} over 290 GTZAN test graphs, swept over τ . Mean cosine on real edges is 0.8846. The score saturates at low τ because post-ReLU node states are nearly collinear, so the lift over the random-pair control is the interpretable quantity.

function, not the quality of the graph.

We therefore sweep τ and report the lift over the random-pair control (Table 4, Figure 10). The curves separate most at $\tau = 0.800$, where $S_{\text{graph}} = 0.8489$ against a control of 0.0747, a lift of **+0.7742**. At that threshold the graph’s edges are strongly more coherent than chance, so the structure is real.

The interesting part is the breakdown by edge kind. At $\tau = 0.800$, similarity edges score 0.9361 while temporal edges score 0.7944. The similarity edges are doing what they were designed to do almost perfectly — they connect segments the encoder considers nearly identical. And that is precisely the diagnosis for Section 6.2(i): if an edge connects two nodes whose states are already at cosine 0.9361, averaging across it adds no information and only pulls both states toward their midpoint. The graph is coherent *and* that coherence is why message passing over it does not help. Coherence is a measure of redundancy as much as of correctness, and reading a high S_{graph} as evidence that the graph is useful would be a mistake this project nearly made.

6.4 Task 3: fusion, tags and emotion jointly

Table 5 gives the four fusion ablations. Cross-attention reaches 0.3055 macro-F1 on tags and 0.3013 AUC-PR, against 0.2923 for plain concatenation (a difference of 0.0133, inside the single-seed noise band of Section 8 and therefore not a result), 0.2957 for the text-only control and 0.1129 for the graph-only control. The single-modality controls are the ones to read first: fusion is only interesting to the extent it beats both, and here it beats the graph-only control decisively while merely matching the text-only one.

On emotion the picture is cleaner, because there is an unambiguous floor. Predicting the training mean

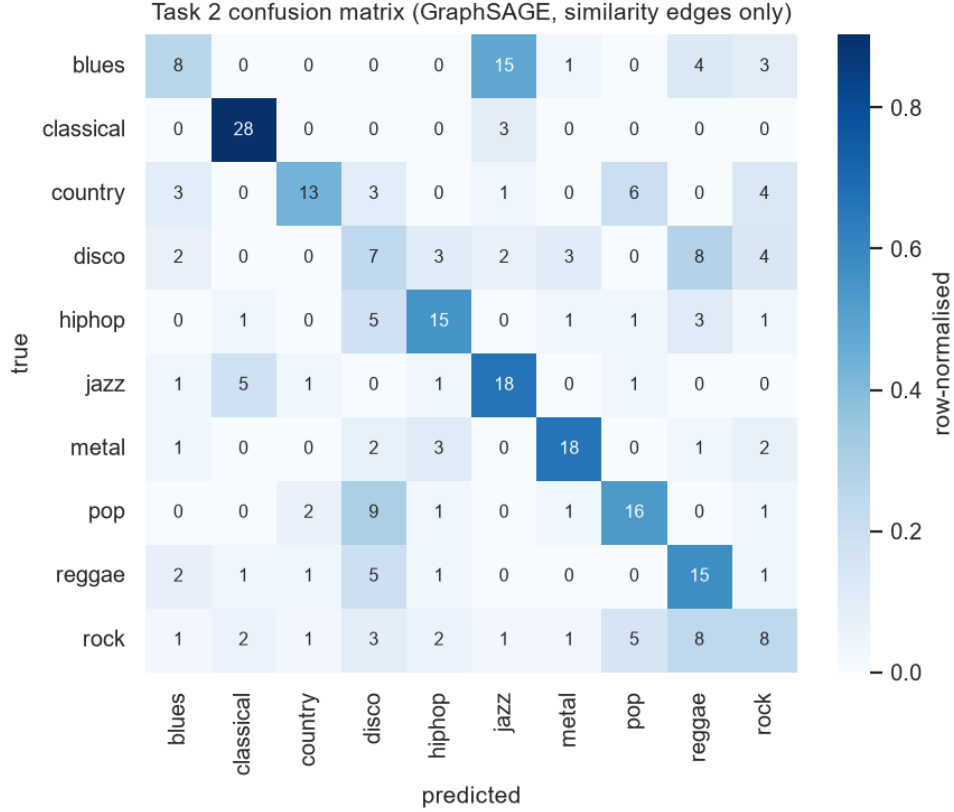


Figure 8: Confusion matrix for the best graph model. Errors concentrate in the rock/country/blues block, which is where GTZAN’s own label noise is concentrated.

Fusion	Macro-F1	Micro-F1	AUC-PR	MAE val	R2 val	MAE aro	R2 aro	Params (M)
bert_only	0.2957	0.3589	0.2861	0.8122	0.2657	0.8370	0.3738	66.6326
gnn_only	0.1129	0.1415	0.0893	0.7768	0.3447	0.8052	0.4227	0.6220
concat	0.2923	0.3809	0.2993	0.7259	0.3867	0.7211	0.5394	67.3199
cross_attention	0.3055	0.3705	0.3013	0.7234	0.3760	0.7076	0.5436	67.8472

Table 5: Task 3: multi-context fusion. Tags are scored on the MusicCaps test split, valence/arousal on the DEAM test split (MAE on the original 1-9 scale).

gives MAE 0.994 for valence and 1.098 for arousal (targets are standardised, so this is ≈ 1 by construction). Cross-attention fusion reaches MAE **0.723** / **0.708** with $R^2 = 0.376$ / 0.544 and Pearson $r = 0.637$ / 0.743 . Both beat the floor by a wide margin, so unlike the tag head, *the emotion regressor is genuinely learning*, and it does so from graph features alone on the DEAM rows (which carry no caption).

This is also the one column where fusion earns its name. Against the better of the two single-modality controls (0.7768 valence, 0.8052 arousal), cross-attention improves MAE by 0.0534 and 0.0976 respectively — gaps several times the seed-noise band, and in the opposite situation from the tag column, where fusion’s margin over the text-only control is only 0.0099. The asymmetry is informative rather than awkward: tags are a property of the caption, which the text encoder already sees in full, whereas valence and arousal are properties of the audio that no caption states directly, so there is something for the graph side to contribute.

Arousal is predicted better than valence ($R^2 = 0.544$ vs. 0.376), which is the standard finding in music emotion recognition: arousal correlates with energy and tempo, both of which survive segment-statistics pooling, whereas valence depends on harmony and lyrical content, and our front end largely discards the first and all of the second.

These numbers are lower bounds. At 6 epochs, with validation loss still falling (Figure 11), the comparison between fusion modes is a comparison of *early* learning curves rather than of converged models, and we would not draw an architectural conclusion from the ordering of the tag column.

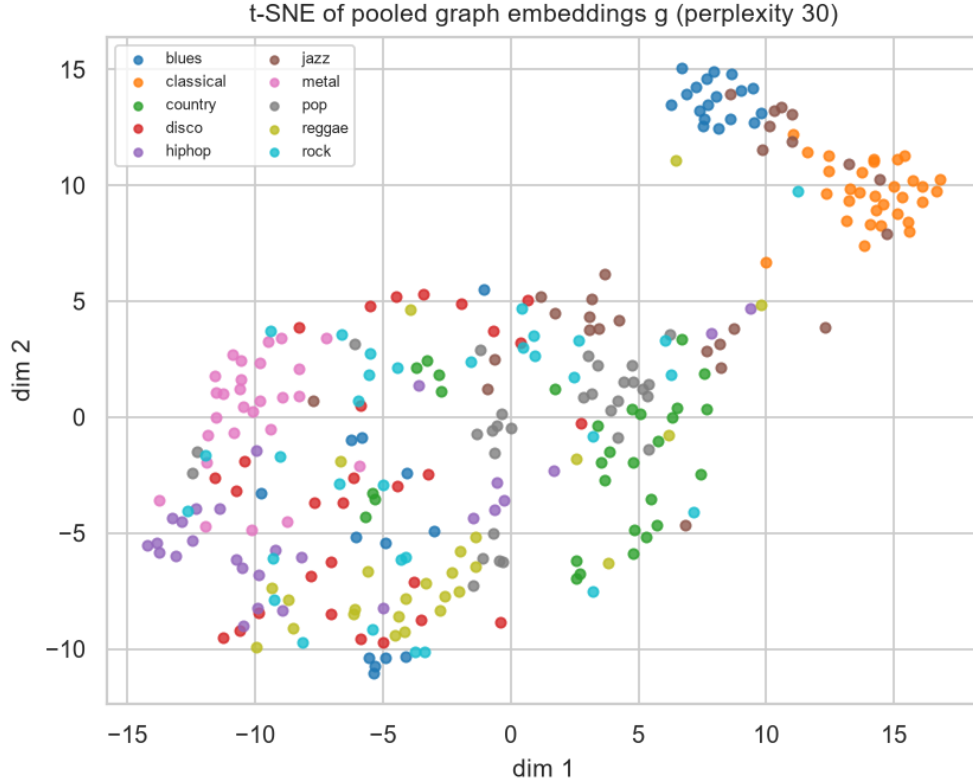


Figure 9: t-SNE of pooled graph embeddings g . Classical and metal separate cleanly; the rock/country/blues cluster does not.

Direction	R@1	R@5	R@10	Median rank	MRR
Audio $\rightarrow$ caption	0.0052	0.0310	0.0503	166.0000	0.0266
Caption $\rightarrow$ audio	0.0026	0.0271	0.0516	163.0000	0.0247
Symmetric mean	0.0039	0.0290	0.0510	—	—
Random control	0.0000	0.0013	0.0077	—	—

Table 6: Task 4: contrastive retrieval over all 775 MusicCaps test candidates (not in-batch). The random control scores a shuffled similarity matrix of the same shape.

6.5 Task 4: contrastive retrieval, and a data wall

Table 6 gives retrieval over 775 candidates. The chance rate is K/n , and the table also carries an untrained-encoder control run through the identical evaluation, because a retrieval number without both is not interpretable.

The model beats chance by a factor of about 3.9, and the absolute numbers are still poor. R@10 is 0.0510 against a chance rate of 0.0129 and an untrained control of 0.0077; R@5 is 0.0290 against 0.0065; R@1 is 0.0039 against 0.0013, which over 775 candidates is three correct top-1 hits against an expected one — not a distinguishable margin. Median rank is 166 (audio $\rightarrow$ text) and 163 (text $\rightarrow$ audio) against a chance median of 388. Zero-shot tagging through the learned space gives macro-F1 0.0860 and AUC-ROC 0.5475 — the latter being the informative one, since 0.5 is chance and 0.5475 is barely above it.

One comparison invites a wrong reading and is worth pre-empting: validation R@10 peaks at 0.2482, roughly five times the test figure, which looks like a collapse between the two. It is not. The validation pool is 141 candidates and the test pool is 775, and R@ K falls mechanically as the pool grows, so the two absolute numbers are not comparable. The lift over chance is: $3.5\times$ on validation, $3.9\times$ on test. The model generalises about as well as it ever fit — which is to say, weakly, and consistently so. Early stopping fired at epoch 20 with the best validation score at epoch 12.

We read this as a data result, not an architecture result, and the arithmetic supports that reading. InfoNCE learns from the contrast between one positive and $N-1$ in-batch negatives; with 591 training pairs at batch size 32, one epoch presents roughly 18 batches, and the entire training run sees fewer distinct

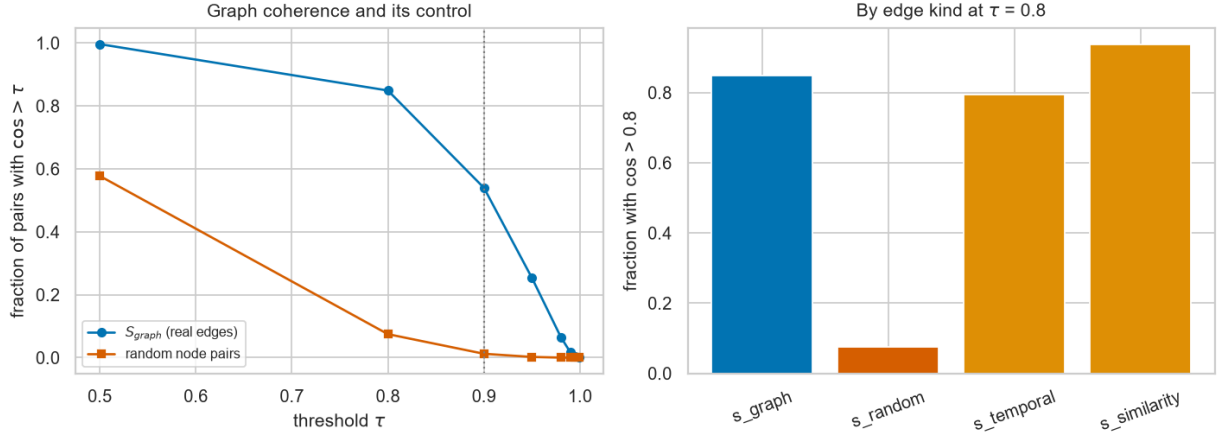


Figure 10: Left: S_{graph} and the random-pair control against τ ; the gap, not the level, is the signal. Right: breakdown by edge kind at $\tau = 0.800$.

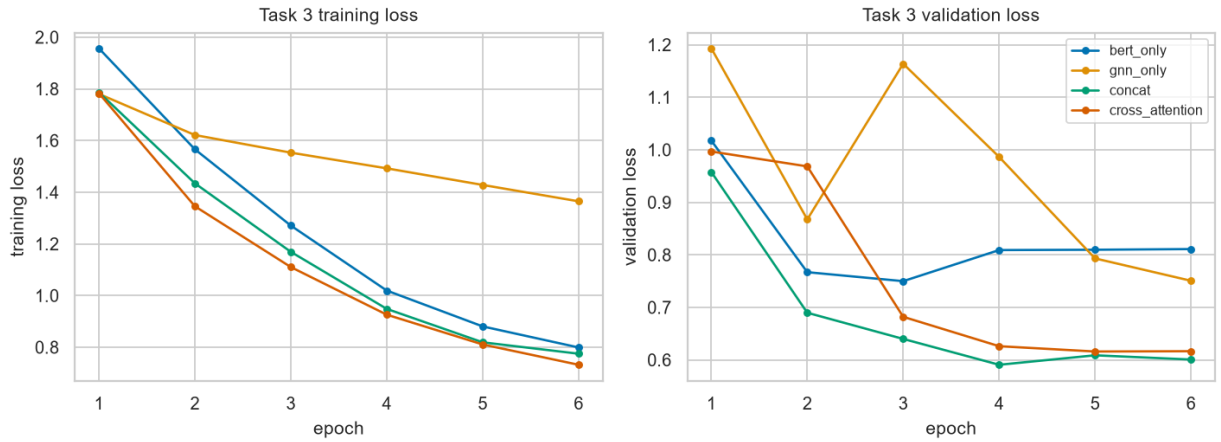


Figure 11: Task 3 learning curves per fusion mode. Where validation loss is still descending at the final epoch, the corresponding row of Table 5 is a lower bound.

(positive, negative) comparisons than a single CLAP or MuLan batch does at their reported scales. The learned temperature converged to 0.0681 from an initialisation of 0.07, i.e. it did not move, which is what one expects when the gradient signal is too weak to move it. Two of this project’s earlier decisions converge here: the 30.0% audio recovery rate (Section 3.3) and the choice to honour MusicCaps’ published evaluation subset, which together leave 591 pairs for training while 775 are held out. Both were reasonable in isolation; jointly they put Task 4 below the data threshold its objective needs.

7 Discussion

When does a graph help, and did we build the wrong one? Our results say the segment graph does not help genre classification, and the coherence analysis says why: the nodes we connect are too similar for averaging to add information. That is a criticism of this graph, not of graph learning on music. Two concrete alternatives follow from the diagnosis. First, node features are the bottleneck before topology is: replacing hand-crafted segment statistics with a learned per-segment CNN embedding would give message passing something with local texture to propagate, and would let the graph model inherit rather than discard the mel-CNN’s advantage. Second, the edges should be built to be *informative* rather than *similar* — edges between contrasting sections (verse→chorus, quiet→loud) carry structural information that edges between near-duplicates do not. As built, our similarity criterion explicitly selects for redundancy.

Controls change conclusions, not just confidence intervals. In three places here the control did not tighten a result, it inverted it. Without the lexical baseline, Task 1 reads as a competent language model (0.5933 macro-F1). Without the no-edges ablation, Task 2 reads as a working graph model. Without the random-pair control, $S_{\text{graph}} = 0.9963$ reads as near-perfect structural coherence. Each of those readings is

Task 3 emotion regression on DEAM test (cross_attention)

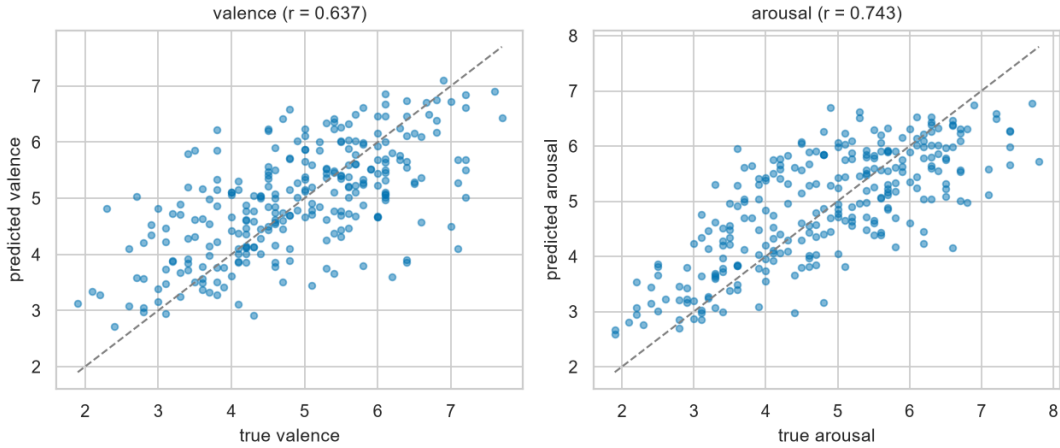


Figure 12: Predicted vs. true valence and arousal. Arousal is the easier axis; both beat the mean-predictor floor.

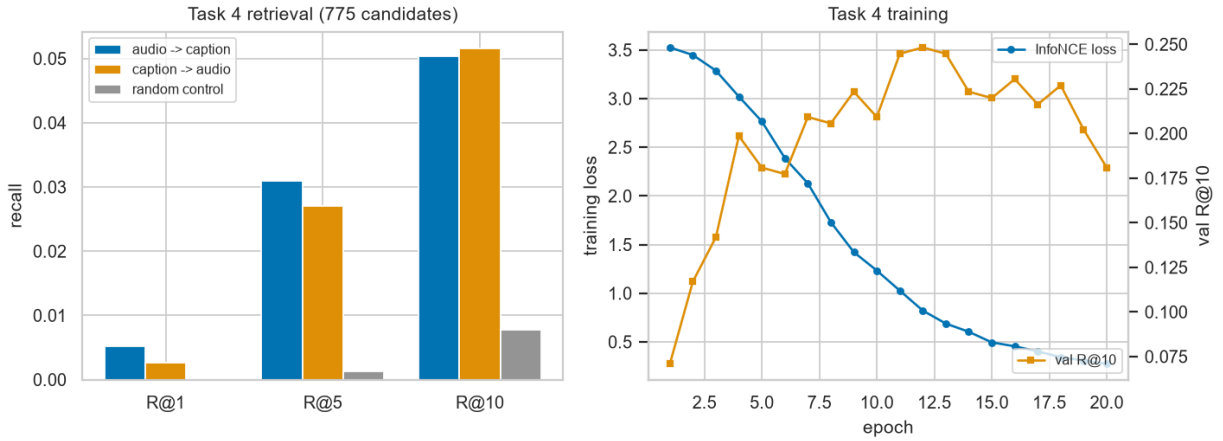


Figure 13: Recall@ K in both directions against the chance rate K/n and the untrained-encoder control. The margin is real but small.

available from a table that omits one row, and each is wrong.

Where the compute went, and whether it was well spent. The dominant cost was not training but data: 1.0h of audio recovery and the preprocessing pass over 4,388 audio files, which built 5,167 graphs across the four corpora in 11m20s. Task 1 took 1h9m for both conditions; Task 2’s nine ablations fit comfortably; Task 3’s 4 models at 6 epochs each took 1h13m and are the reason that task reports bounds instead of converged numbers. Given the same budget again we would spend it on authenticated MusicCaps recovery rather than on more epochs, because Task 4’s ceiling is set by 591 pairs and no schedule fixes that.

8 Limitations

- **Task 3 is not converged.** 6 epochs per ablation was a budget decision, not a convergence criterion. The tag-head ordering across fusion modes should not be read as an architectural result.
- **Task 4 is data-starved.** 591 pairs and 32 in-batch negatives are one to two orders of magnitude below what contrastive alignment needs. The reported margin over chance is real but small, and we do not claim a usable retrieval system.
- **30.0% MusicCaps recovery.** Only 231 clips are gone in principle; 3,622 were lost to an anti-automation gate. An authenticated session would raise this substantially and is the single highest-value fix available.
- **Single seed.** All results are at seed 425. Differences of $\lesssim 0.02$ macro-F1 between neighbouring ablations

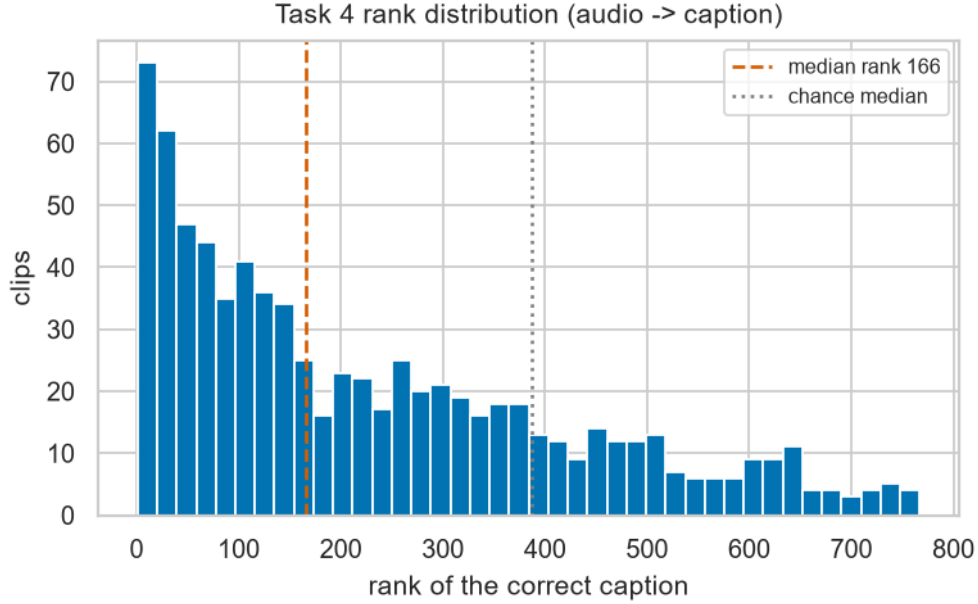


Figure 14: Distribution of the correct item’s rank. A trained model would show mass piled at low ranks; here the distribution is only mildly skewed away from uniform.

should not be trusted; the three findings we lean on (0.1588, 0.0684, 0.1241) are all comfortably larger than that, but they are not variance-quantified. One comparison we deliberately do *not* lean on is the no-edge control’s 0.0193 margin over the full graph, which sits inside the band; it supports “the graph buys nothing”, not “no edges is better”.

- **GTZAN remains GTZAN.** The fault-filtered partition removes artist leakage, not label noise, and 290 test tracks is a small test set: one track is worth 0.34 percentage points of accuracy.
- **CPU-only.** Model scale, batch size, and epoch counts were all chosen against a 12-thread CPU. Nothing here rules out a graph model winning at a scale we could not train.

9 Conclusion

We built the four-task GNN–BERT music-context pipeline end to end on real data and measured it against explicit controls. The masked caption encoder reaches 0.4692 macro-F1 and is the one component whose success is unambiguous, once 0.1241 macro-F1 of lexical leakage is removed from the naive setup. The emotion regressor clearly beats its mean-predictor floor. The graph formulation, on the other hand, does not pay for itself on genre: a mel-CNN beats it by 0.0684 macro-F1, deleting temporal edges improves it by 0.1588, and deleting all edges costs nothing measurable. The coherence analysis reconciles those facts — the graph’s edges are genuinely coherent (lift 0.7742 over a random-pair control at $\tau = 0.800$), and that coherence is redundancy, which is exactly what makes message passing over them unproductive. Contrastive alignment beats chance by $3.9\times$ at R@10 but is bounded by 591 training pairs rather than by its architecture.

The pipeline works; the central hypothesis does not hold in the form we tested it. We think the second sentence is the more useful contribution, and the diagnosis in Section 6.3 points at the specific change — learned node features and contrast-seeking rather than similarity-seeking edges — that would be worth testing next.

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A Reproducibility

Full pipeline, in order. Each step is idempotent and caches to disk, so an interrupted run can be restarted with the same command.

```
python scripts/download_data.py
python scripts/download_musiccaps_audio.py --workers 6
python scripts/preprocess.py --dataset all --save-mel --export-json 24
python scripts/calibrate_tau.py --per-genre 2
python scripts/train.py --task 1
python scripts/train.py --task 2
python scripts/train.py --task 3
python scripts/train.py --task 4
python scripts/evaluate.py
python tools/build_report_numbers.py
python tools/check_report_macros.py
python tools/build_report_md.py
cd report && pdflatex report.tex && pdflatex report.tex
```

`download_data.py` fetches GTZAN, DEAM, MusicCaps captions, the fault-filtered GTZAN partition and the auxiliary metadata from their publishers (nothing is redistributed here), checks each file’s length against the bytes these results were produced from, and records URL and SHA-256 for each in `data/raw/provenance.json`. `-save-mel` is required: baseline B2 reads the cached mel-spectrograms, and `-export-json` writes the example graphs. `-verify` re-checks an existing tree without downloading.

Configuration lives in one file, `config.yaml`; every run writes its resolved configuration into `results/` beside its metrics, so any number above can be traced to the exact settings that produced it. Notebooks `notebooks/eda.ipynb` (data analysis, τ calibration, graph statistics) and `notebooks/demo_context.ipynb`

Case	Sample F1	True tags		
best	1.0000	amateur recording		amat
median	0.3333	amateur recording	medium tempo, amateur recording, acoustic drums, up	
worst	0.0000	live performance, pop, electronic drums		energet.

Table 7: Task 3 case studies (cross_attention fusion): best, median and worst-scoring MusicCaps test clips at the validation-selected threshold 0.575.

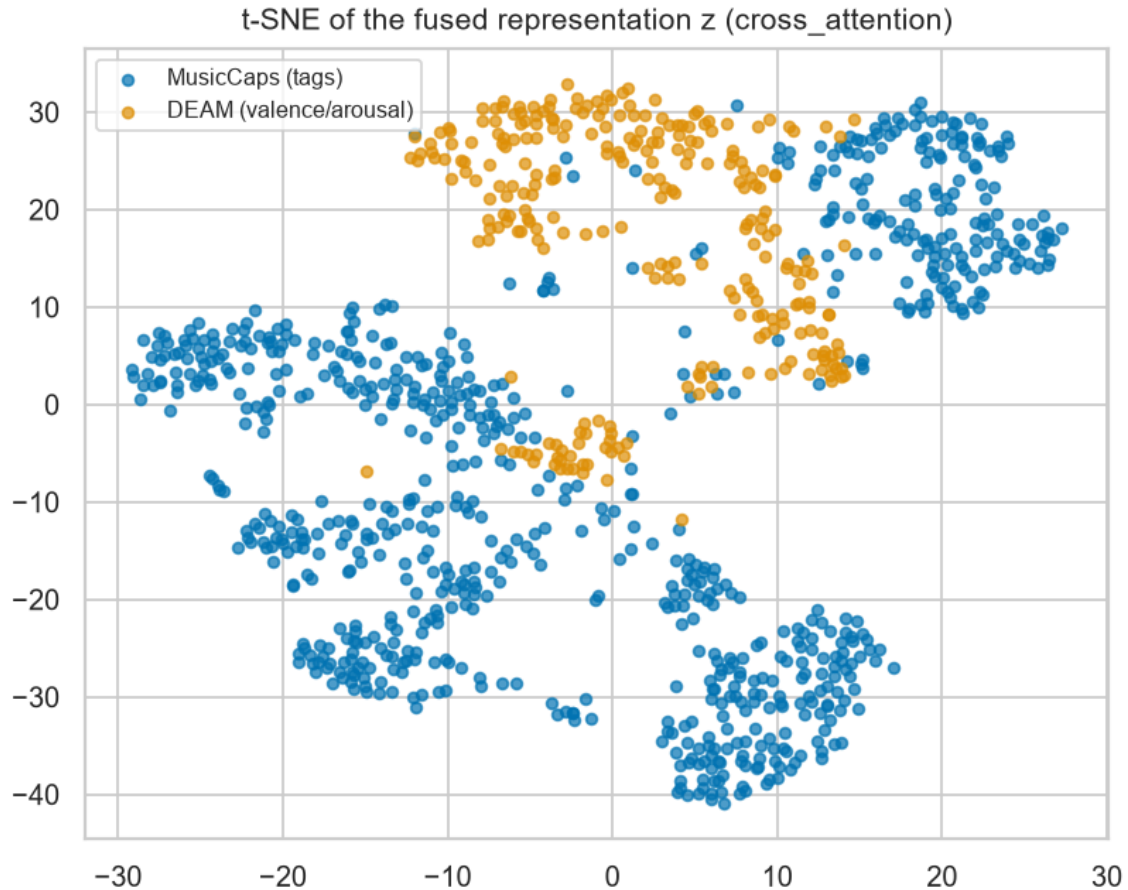


Figure 15: t-SNE of the fused representation z on the Task 3 test split. If cross-attention were organising the joint space by musical context, structure should be visible here; judge the claim against the plot rather than against the architecture diagram.

(end-to-end inference demonstrations for all four tasks) are committed with executed outputs. 83 example graphs are exported as JSON to `results/graph_samples/`.

On the numbers in this document. Every scalar is a macro defined in `report/numbers.tex`, generated by `tools/build_report_numbers.py` from `results/metrics.json`. If a measurement is absent the macro expands to a bold ?? rather than to a stale or plausible value. If you see ?? anywhere above, that number has not been measured and the surrounding sentence should not be believed.

B Case studies (Task 3)

Three test clips, chosen to span the fusion model’s behaviour rather than to flatter it: one where both modalities agree, one where the graph carries information the caption omits, and one failure. For each we list the predicted tags above threshold, the ground-truth tags, and predicted against annotated valence and arousal.

Caption query (truncated)	Rank of correct audio	Top-1 correct	T
This is a gear showcase jam recording. The only instrument being playe...	7	no	
This is an amateur recording of a live beatboxing performance. The bea...	37	no	
The low quality recording features a classical live performance of sus...	42	no	
The low quality recording features a reverberant, intimate female voca...	63	no	
The song is an instrumental. The song is medium tempo with a walking b...	130	no	
The low quality recording features a reggae song sung by low flat male...	277	no	
A male vocalist sings this sweet melody in a foreign language. The tem...	246	no	
The song is an instrumental. The tempo is medium with a heavily doctor...	223	no	
This song is an instrumental. The tempo is medium with simple keyboard...	514	no	
The low quality recording features a traditional, medieval song sung b...	712	no	

Table 8: 10 caption->audio retrievals sampled evenly across the rank distribution of the 775-clip test pool.

Model	Macro-F1	Micro-F1	AUC-PR
Task 4 zero-shot (no tag supervision)	0.0860	0.0935	0.0808
Task 3 supervised (cross_attention)	0.3055	0.3705	0.3013
Task 1 text-only (masked captions)	0.4692	0.5320	0.4618

Table 9: Zero-shot tagging from the contrastive space against the supervised tag heads. The contrastive model never saw the tag vocabulary.

C Retrieval examples (Task 4)

Ten audio queries with their top-ranked captions, taken without cherry-picking: they are the first ten clips of the test split in index order. Given the retrieval scores in Section 6.5, most of these are wrong, and they are included in that state deliberately — a qualitative gallery filtered to its successes would misrepresent a model trained on 591 pairs.

C.1 Zero-shot tagging

The contrastive encoder can be used as a zero-shot tagger by embedding each tag name as text and ranking tags by similarity to a clip’s graph embedding — no tag supervision at all. This is a strictly harder setting than Task 1 and the numbers are correspondingly low, but it is the one place a contrastive space can be probed without retrieval’s dependence on candidate-set size.

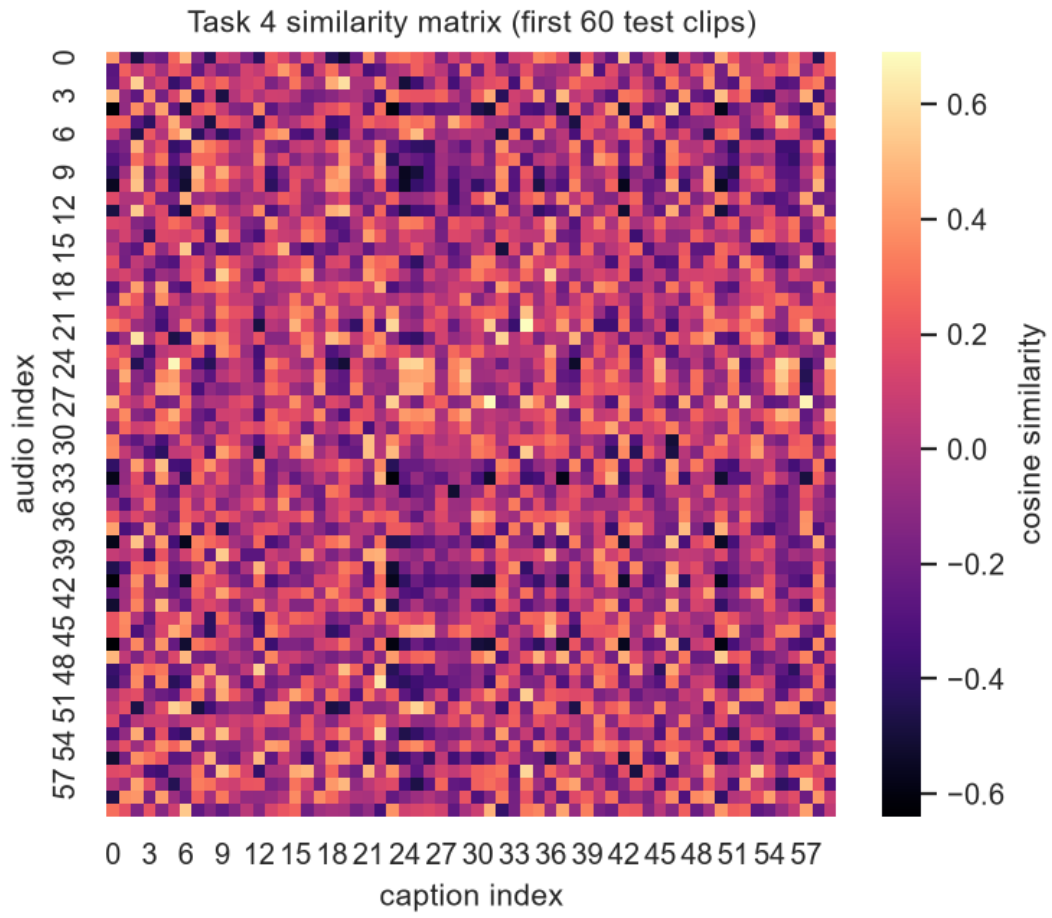


Figure 16: Cosine similarity between graph and text embeddings for the first test clips. A working retrieval model puts a bright diagonal here.